

✓ PDS PRACTICAL 4

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- **Roll no: 34**
- **Batch: 02**
- **Date of Assignment: 19/08/26**
- **Date of Submission: 19/08/26**

```
import pandas as pd
```

6A. Write a Python Program to import a CSV Dataset using Pandas and display the first 5 and Last 5 records

```
df= pd.read_csv('Students.csv')
```

```
print(df.head())
```

	Roll No	Name	Department	Marks	Attendance
0	101	Amit	BSc IT	78	85
1	102	Priya	BSc CS	65	90
2	103	Rahul	BSc IT	88	92
3	104	Neha	BSc CS	72	80
4	105	Rohan	BSc IT	55	70

```
print(df.tail())
```

	Roll No	Name	Department	Marks	Attendance
0	101	Amit	BSc IT	78	85
1	102	Priya	BSc CS	65	90
2	103	Rahul	BSc IT	88	92
3	104	Neha	BSc CS	72	80
4	105	Rohan	BSc IT	55	70

6B. Write a Python Program to analyze the dataset by displaying the number of rows and columns, column names, data types, and basic statistical information

```
print(df.shape)
```

```
(5, 5)
```

```
print(df.dtypes)
```

```
Roll No      int64
Name         object
Department   object
Marks        int64
Attendance   int64
dtype: object
```

```
print(df.columns)
```

```
Index(['Roll No', 'Name', 'Department', 'Marks', 'Attendance'], dtype='object')
```

```
print(df.describe)
```

```
<bound method NDFrame.describe of
0      101  Amit    BSc IT    78      85
1      102  Priya   BSc CS    65      90
2      103  Rahul   BSc IT    88      92
3      104  Neha    BSc CS    72      80
4      105  Rohan   BSc IT    55      70>
```

```
result=df[df["Attendance"]>=60]
print(result)
```

```
Roll No  Name Department  Marks  Attendance
0      101  Amit    BSc IT    78      85
1      102  Priya   BSc CS    65      90
2      103  Rahul   BSc IT    88      92
3      104  Neha    BSc CS    72      80
4      105  Rohan   BSc IT    55      70
```

6C. Write Python Program to filter records from the dataset using conditions. For example, Display studnets having marks greater than 60

```
result=df.sort_values("Marks")
print(result)
```

	Roll No	Name	Department	Marks	Attendance
4	105	Rohan	BSc IT	55	70
1	102	Priya	BSc CS	65	90
3	104	Neha	BSc CS	72	80
0	101	Amit	BSc IT	78	85
2	103	Rahul	BSc IT	88	92

6D. Write a python program to perform data anaylsis on the dataset by finding the maximum, minimum, average and total values of a numeric column

```
print("Minimum Marks= ", df["Marks"].min())
```

Minimum Marks= 55

```
print("Maximum Marks= ", df["Marks"].max())
```

Maximum Marks= 88

```
print("Average Marks= ", df["Marks"].mean())
```

Average Marks= 71.6